RAGURAM KASTHURISAMY

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EDUCATION

Northeastern University, Boston, MA | *Master of Science in Electrical and Computer Engineering* *Expected May 2028*
Relevant Coursework: Computer Architecture, Computer Systems

Kumaraguru College of Technology, Coimbatore, India | *Bachelor of Engineering in Mechatronics Engineering* *May 2024*
Relevant Coursework: Digital Electronics & Microprocessors, Embedded Systems, Sensors & Instrumentation, Industrial Electronics & Drives, Advanced IoT

EXPERIENCE

Northeastern Electric Racing, Boston, MA | *Firmware Team Member* *Sep 2026 - Present*
- Completed NER's Firmware Launchpad onboarding program, building an STM32-based mini car starter project in embedded Rust with the Embassy async framework, driving GPIO, ADC, and I2C peripherals
- Advanced to race car development, writing Rust/Embassy firmware for the electric race car's STM32-based boards

Soliton Technologies, Coimbatore, India
Project Engineer (Contract), National Instruments *Jan 2026 - Aug 2026*
- Built an auto-scheduling algorithm for NI's lab-management software that assigns test plans to available test systems, replacing ~1 min of manual effort per item for labs scheduling 500-1,000 work items daily
- Stabilized 1,000-1,200 intermittently failing UI tests by replacing hard-coded timeouts with dynamic asynchronous waits

Intern, Protocol Validation Suite *Jan 2024 - Aug 2024*
- Expanded Soliton's I2C-only Protocol Validation Suite (PVS) to MIPI I3C by developing LabVIEW functional-compliance tests against the I3C specification, helping launch an I3C validation product sold through NI and showcased at MIPI Alliance Demo Day
- Developed parametric tests measuring I3C bus timing and electrical levels against spec limits on NI PXIe-657x digital pattern instruments, giving silicon vendors pass/fail compliance data for their devices
- Built PVS framework APIs with TestStand/Python integration so chip vendors could run I3C compliance suites from existing automation

KLA, Chennai, India | *Project Engineer (Contract)* *Jan 2025 - Dec 2025*
- Rebuilt legacy C++ MFC recipe editor in C#/.NET Blazor and migrated result viewer to C#/WPF for Archer 800 and ATL 200 overlay metrology tools, cutting startup time and memory use for a leading semiconductor foundry's wafer-inspection workflow
- Introduced SonarQube to KLA's Optical Metrology Division across ~1,700+ C#/C++ projects in Jenkins CI/CD, surfacing and resolving ~560+ defects; fixing all 360 undisposed resources it flagged cut application memory leakage by 50%
- Integrated Squish Coco code coverage, new to KLA, into 1,700+ C#/C++ project builds, automating previously manual coverage reports

iQube Innovation Forum, Kumaraguru College of Technology, Coimbatore, India | *Embedded & IoT Intern* *Mar 2021 - Dec 2023*
- Eliminated 3-5 mm positioning drift in a DD Robocon India 2022 robot's open-loop DC drive by proposing encoder-feedback motors and writing STM32 PID control firmware, helping the team clear the first two rounds of the national qualifier for ABU Robocon
- Developed battery management firmware for the Robocon robot's multi-cell Li-ion pack using BMS ICs, current sensing, and ADC measurements, tracking state of charge, battery health, and charge/discharge cycles to predict remaining runtime
- Centralized lab control with an MQTT/Wi-Fi network of 10 ESP32/ESP8266 nodes driving relays for 40-50 appliances and 3 room doors
- Owned electrical wiring and power distribution for the Robocon robot across 15+ PCBs, 20-25 sensors, and 10+ actuators, using oscilloscopes, logic analyzers, and bench testing to isolate faults and speed up team debugging

PROJECTS

Bare-Metal UART Driver Study, STM32U575 | *github.com/Ragu-programmer/Device-Driver-Development-STM32-MCU* *Oct 2026*
- Investigated how data-transfer strategy affects CPU load and data integrity on a Cortex-M33 by designing, implementing, and benchmarking three bare-metal USART drivers without HAL: polling, interrupt-driven, and DMA-based
- Built each tier at register level from the RM0456 reference manual, configuring RCC clocks, GPIO alternate functions, NVIC priorities, and GPDMA channels, with lock-free ring buffers for interrupt I/O and circular GPDMA reception
- Benchmarked all three tiers on hardware, confirming CPU availability improved from polling to interrupts to DMA, and cut CPU interrupts per 100-byte transfer from 100 to 1 with GPDMA, identifying DMA as the best fit for high-throughput streams

STM32-based Robot Actuator Controller | *github.com/Ragu-programmer/STM-32_Motor_Controller* *Jun 2023 - Dec 2023*
- Built an STM32F446RE gripper controller with PWM drive, quadrature encoders, UART commands, and interrupt-based position updates
- Transitioned from time-based open-loop actuation to encoder-feedback closed-loop position control, improving positioning consistency by ~5× and reducing motion variation from ~17% to within ±2%
- Tuned practical PD control with startup boost, dead zone compensation, velocity-aware arrival detection, software limits, and homing-settle calibration to handle actuator friction, inertia, and backlash

Custom STM32 Bootloader | *github.com/Ragu-programmer/Custom-Bootloader-for-STM-32-MCU* *May 2023 - Oct 2023*
- Developed an STM32 bootloader in C/C++ for in-field firmware updates over UART/SPI, with flash partitioning and safe jump-to-application and designed a packet-based transfer protocol with error handling and retransmission for reliable firmware delivery
- Implemented CRC/checksum validation and a fault-tolerant recovery state machine to prevent bricking on failed or corrupted updates

TECHNICAL SKILLS

Programming Languages	Embedded C, C++, Assembly, Python, Rust, C#/.NET, Shell scripting
MCUs & Architectures	ARM Cortex-M (STM32), RISC-V, AVR, PIC32, ESP32/ESP8266, Arduino, Raspberry Pi
Firmware & RTOS	Bare-metal, Embassy, Zephyr RTOS, nRF Connect SDK, Device Drivers, Bootloaders, Interrupts & NVIC, DMA
Protocols & Peripherals	UART, SPI, I2C, MIPI I3C, MQTT, Wi-Fi; ADC/DAC, Timers, PWM, GPIO
Hardware & Debug	Hardware bring-up, PCB design (KiCad, Altium), Oscilloscopes, Logic analyzers, Sensor & actuator integration
Testing & Quality	Ceedling, Unity, CMock, GoogleTest, Cppcheck, SonarQube, LabVIEW, TestStand
Developer Tools	STM32CubeIDE, Git/GitHub, CMake, Jenkins CI/CD, Jira, Linux